

Lasso

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Prerequisites: Linear regression.

We define and process $\mathbf{X}, \mathbf{y}, \mathbf{w}$ similar to linear regression.

Lasso regression extends linear regression by adding an L_1 regularization term to the loss function. The resulting optimization problem is

$$\hat{\mathbf{w}} = \underset{\mathbf{w}}{\operatorname{argmin}} \left(\|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2 + \lambda \|\mathbf{w}\|_1 \right),$$

where

$$\|\mathbf{w}\|_1 = \sum_{j=1}^p |w_j|$$

is the L_1 norm of the coefficient vector, and $\lambda \geq 0$ is the regularization parameter.

Unlike ordinary linear regression, the Lasso objective function is not differentiable at $w_j = 0$ because of the absolute value term in the L_1 norm. Consequently, there is no closed-form solution for $\hat{\mathbf{w}}$.

Instead, iterative optimization algorithms are used to compute the solution. One of the most common approaches is coordinate descent, which updates one coefficient at a time while keeping all other coefficients fixed.

For a coefficient w_j , define the partial residual

$$r_i^{(j)} = y_i - \sum_{k \neq j} x_{ik} w_k.$$

Substituting this residual into the objective function gives

$$\min_{w_j} \sum_{i=1}^n \left(r_i^{(j)} - x_{ij} w_j \right)^2 + \lambda |w_j|.$$

Define

$$\rho_j = \sum_{i=1}^n x_{ij} r_i^{(j)},$$

and

$$z_j = \sum_{i=1}^n x_{ij}^2.$$

The optimal update for w_j is given by the soft-thresholding operator

$$w_j = \frac{S(\rho_j, \lambda/2)}{z_j},$$

where

$$S(\rho, \gamma) = \begin{cases} \rho - \gamma, & \rho > \gamma, \\ 0, & |\rho| \leq \gamma, \\ \rho + \gamma, & \rho < -\gamma. \end{cases}$$

The coefficients are updated sequentially using this rule until the change in the coefficient vector becomes sufficiently small.